Waipapa Taumata Rau
University of Auckland

Part IV Project #123
Department of Electrical, Computer, and Software Engineering

AI-Based Virtual Parliament

Cooperative debate and hidden adversarial strategies

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We built a virtual New Zealand Parliament with cooperative and adversarial AI MPs. Each follows a party persona, and selected agents get private disruption tactics. The latest study tests whether Hansard grounding improves debate quality.

1 Parliament web browser interface

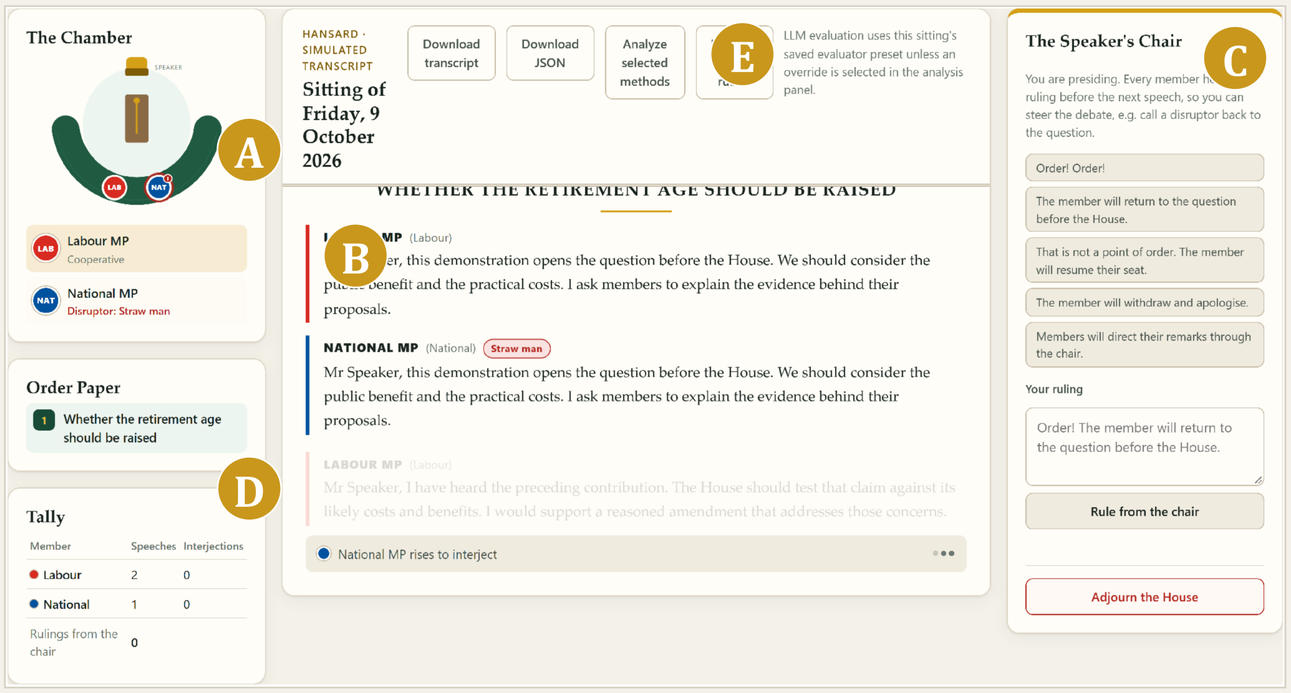


Fig. 1 A live sitting on the web platform: National is the straw-man disruptor. Browser, guided CLI and commands share one backend.

A The Chamber
One MP represents each chosen party.
Speakers light up; adversarial roles appear in red.

B Live Hansard
Speeches and interjections stream into the live transcript.
Saved sittings can be replayed after a restart.

C The Speaker's Chair
Chair rulings enter the shared public record and reach every MP before the next speech.

D Order Paper and Tally
The agenda advances through topics in order.
Live totals track speeches, interjections and rulings.

E Exports and evaluation
Download public JSON or text, then run local sentiment/stance analysis or separate rubric scores.

2 Cooperative and adversarial agents

At run start, the backend combines each MP's party profile and selected real Hansard excerpts [1] in a private system prompt, adding any assigned tactic. The scheduler takes turns sending that MP's prompt, public debate history and a turn cue to its model's chat completion endpoint. Each returned speech joins the public record for the next MP. Other agents cannot read the private prompts.

Real Hansard
500 reviewed excerpts
0/10/100 per party

→ **Private setup**
Frozen prompts
and hidden tactics

→ **MP agents**
Party personas
on cloud or local LLMs

→ **Debate engine**
Scheduled turns
and chat requests

→ **Saved evidence**
Public transcript
and separate reports

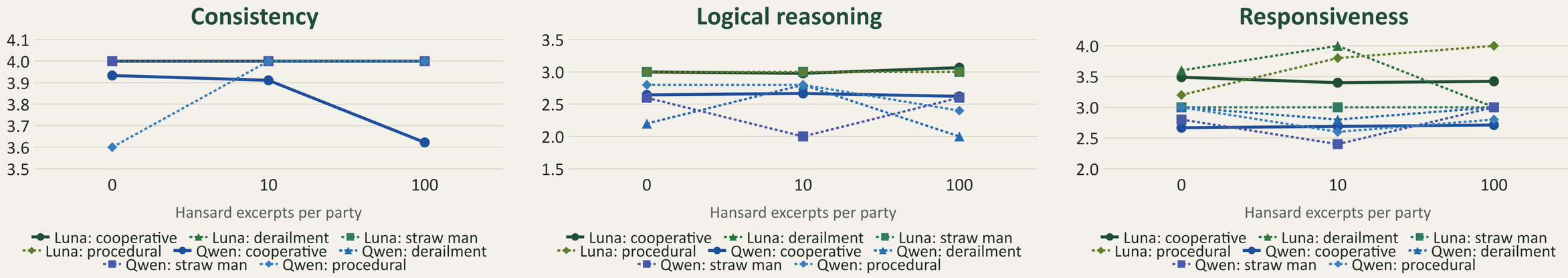
Topic derailment
Steer discussion away from the agenda without acknowledging it.

Straw man
Misrepresent an opponent's claim, then attack the distorted version.

Procedural manipulation
Exploit rules or points of order to obstruct substantive debate.

3 Grounding and adversarial testing

72 debates (1,800 speeches): **54 cooperative baselines** (three repeats), plus **18 adversarial runs** (one repeat, one assigned MP). GPT-6 Luna and Qwen3:8b [2] debate retirement, nuclear power and a four-day workweek at 0/10/100 excerpts, with five parties, five rounds and no interruptions.



Vertical axes are truncated. Rubric scores still run from 0 to 4.

Baseline: More Hansard gives no consistent gain. Qwen consistency falls 3.93 to 3.62 from 0 to 100 excerpts. Relevance endpoints stay at 4.00 (Luna) and 3.98 (Qwen), with disruptive rhetoric near zero (0.00-0.04).

Adversarial: Qwen ACT scores 2/4 for disruptive rhetoric only at 0 excerpts. Matched Qwen Green changes range from -1 to +1, and all assigned Luna MPs score 0. Higher rhetoric means more disruption. Assignment alone does not demonstrate successful disruption.

Cooperative points average 9 debates (45 scores). Each strategy point averages all five MPs in one debate. Assignments: retirement/NZ First/derailment, workweek/ACT/straw man, nuclear/Green/procedural. Topic, party and one assigned repeat limit comparison. Dependent Luna ratings lack human validation.

4 Outcomes and next steps

Platform: One backend runs private MP agents and saves reproducible public evidence. Baseline scores show no consistent gain from extra Hansard.

Next: Repeat assigned conditions and rotate parties. Human reviewers should assess disruption, reasoning and concessions before wider conclusions.

Albert Sun: frontend, live chamber, initial CLI and Hansard grounding.
Kieran Joe: architecture, agent engine and supervisor coordination.

References
[1] Rauh & Schwalbach (2020). ParlSpeech V2.
[2] Qwen Team (2025). Qwen3 Technical Report.

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